

PLUM PROJECT



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Project Title	PLUM Peat and Mangrove Conservation and Restoration Project (PLUM Project)
Version	1.0
Date of Issue	23-February-2024
Project Location	Indonesia, Central Kalimantan
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Project Lifetime	01 August 2022 – 31 July 2082; 60-year lifetime
GHG Accounting Period	01 August 2022 – 31 July 2082; 60-year total period
History of CCB Status	Combined validation with VCS
Gold Level Criteria	Gold Level for Climate Change Adaptation Benefits (GL1), Exceptional Benefits for Community (GL2) and Biodiversity (GL3) GL1: The project will implement activities that increase the resilience of local communities and biodiversity to climate change, including activities for livelihood diversification and habitat conservation and restoration. GL2: A total of 13 villages adjacent to the project area are included as the target beneficiaries of this project, including six villages classified as

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1 SUMMARY OF PROJECT BENEFITS

The PLUM Peat and Mangrove Conservation and Restoration Project (PLUM Project) is a conservation and restoration project of peat and mangrove forests located in Central Kalimantan Province, Indonesia. The project is implemented on a forest concession with Forest Utilization Business License held by PT Pagatan Usaha Makmur (PT PUM) for 60 years, covering a total area of 23,665 hectares. Prior to PT PUM receiving its principal PBPH license in 2022¹, the project area was planned to be developed as an oil palm plantation along with two other companies adjacent to the area.

The project is expected to have a range of climate, community and biodiversity benefits by preserving and restoring the integrity of the Peat Swamp and Mangrove Forest landscape and the rich biodiversity in the area while improving the well-being of local communities. In terms of climate benefits, it is estimated the PLUM Project will result in net emissions reductions and removals of 69,999,940 tCO₂e over 60 years of crediting period. This will be achieved through activities covered by the VM0007 REDD+ Methodology Framework including (1) Avoiding Planned Deforestation (APD), (2) Afforestation, Reforestation and Revegetation (ARR), (3) Restoration of Wetlands Ecosystems (RWE) on Peatland and Tidal Wetlands, and (4) Avoiding Planned Wetland Degradation (APWD) on Peatland and Tidal Wetlands. Additionally, the project will deliver Climate Change Adaptation Benefits by increasing the resilience of local communities and biodiversity to climate change.

The project meets the CCB criteria for Exceptional Community Benefits. There are 13 villages within the project zone and over 50% of villagers are below the national poverty line of IDR 2,120,371/month per household. Furthermore, six of the 13 villages in the project zone are underdeveloped villages. These six villages are the main target of the project's community development programs. The project activities will target the most vulnerable communities within the project zone, improve access to health services and provide fair and just employment opportunities. The project activities will provide a positive impact on communities' livelihoods and well-being.

The project also meets the CCB criteria for Exceptional Biodiversity Benefits. PT PUM's ecosystems are home to 31 globally threatened species with three Critically Endangered (CR) species (i.e., Bornean orangutan), eight Endangered (EN) species (i.e., Proboscis monkey, Bornean white-bearded gibbon), and 20 Vulnerable (VU) species. Furthermore, the project area is home to restricted range species that can only be seen on Kalimantan Island including four endemic mammals and one endemic bird. The project activities will contribute to the conservation of these species and their habitat, and restore the degraded peatland and mangroves to improve the habitat quality and reduce the threats to biodiversity.

1.1 Unique Project Benefits

Outcome or Impact Estimated by the End of Project Lifetime	Section Reference
1) A total of 21,010 ha peatland will be conserved and restored through peat rewetting (e.g., canal blocking) and fire prevention and management activities. This effort will allow degraded areas to recover and hence provide suitable habitats for key species. In addition, negative health impacts from fire smoke and haze to local communities will be reduced.	2.1.11, 3.2.2

¹ Principal license granted to PT PUM by the Ministry of Investment since 1 August 2022 (Decision of the Head of Investment Coordination No.01082211116206001), followed by the definitive license on November 27th, 2023.

2 GENERAL

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (G1.2)

The PLUM Peat and Mangrove Conservation and Restoration Project (PLUM Project) is a conservation and restoration project of peat and mangrove forests located in Central Kalimantan Province, Indonesia. The project is implemented on a 23,665-hectare forest concession with Forest Utilization Business License, held by PT Pagatan Usaha Makmur (PT PUM) for 60 years. The project will implement activities to conserve and restore the peat and mangrove forests in the project area while delivering positive impact on the community and biodiversity. Among others, the project will implement forest patrols, fire prevention, and biodiversity monitoring activities to prevent forest loss and protect biodiversity. Peat ecosystems will be restored by blocking canals and rewetting the area to avoid further peat drainage. Approximately 15,088 ha of degraded land will be restored through reforestation and Assisted Natural Regeneration (ANR).

The eastern part of the project is a peatland forest area located in Katingan Regency, Katingan Kuala Sub-district, adjacent to the Sebangau National Park. The western part of the concession is located in East Kotawaringin Regency, Pulau Hanaut Sub-district, and mostly consists of mangrove forests (Map 2-1). Based on the conducted biodiversity assessment, the concession provides important habitat for 31 threatened species, including the Bornean orangutan, the Proboscis monkey, and the Bornean white-bearded gibbon. The area is also important for at least 17 migratory bird species from three families.

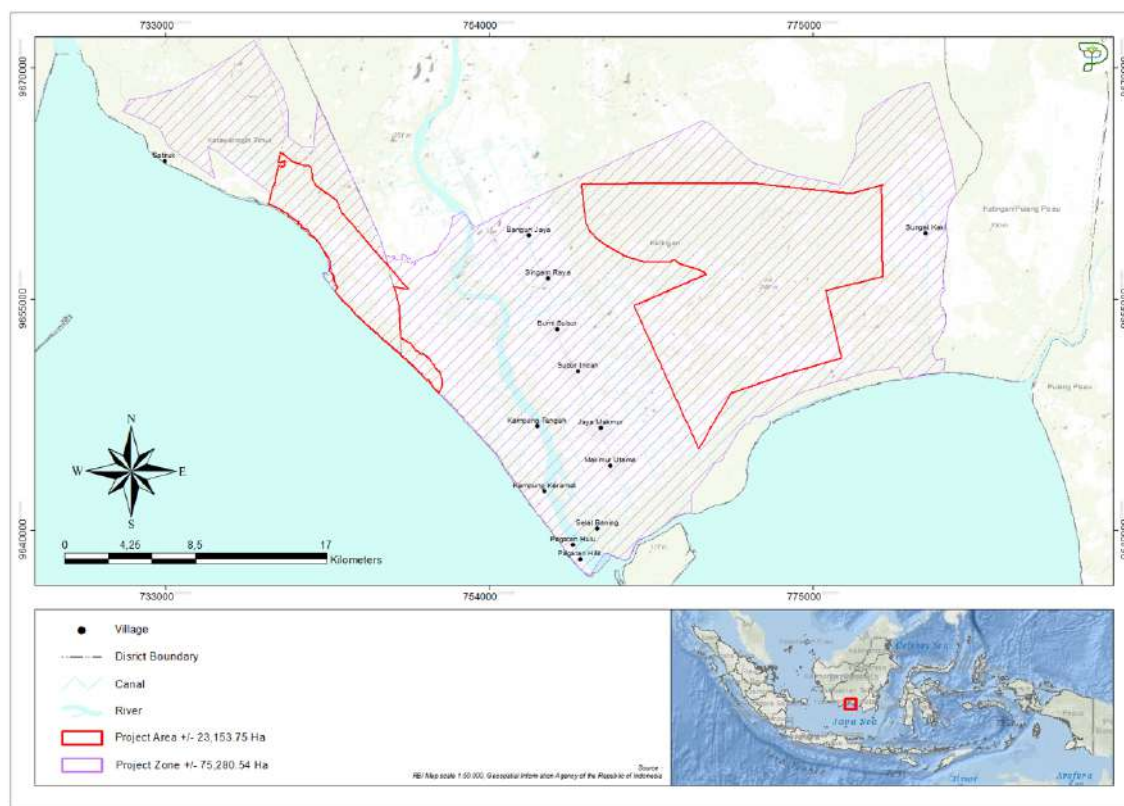
The project will provide tangible benefits to local communities living in the surrounding area by working to improve local livelihoods, health, education, and well-being. For example, the project will assist local coffee farmers and honey producers to improve harvesting, processing, and market access. The project will support the establishment of women's groups in the target villages to promote economic opportunities for women; and implement interventions that improve health and sanitation in the 13 target villages.

Prior to PT PUM receiving its principal PBPH license in 2022, the project area was planned to be developed as an oil palm plantation along with two other companies adjacent to the area. All three companies were owned by a company group named PT Agro Inti Semesta (PT AIS). Additionally, in 2022, an approval to convert forest land into oil palm plantation was given to another oil palm company located along the western boundary of PT PUM. Without the PLUM Project, it is highly likely that PT PUM's area would have been developed into an oil palm plantation by PT AIS.

The project will generate GHG emission reductions or removals by implementing activities covered by the VM0007 REDD+ Methodology Framework including (1) Avoiding Planned Deforestation (APD), (2) Afforestation, Reforestation and Revegetation (ARR), (3) Restoration of Wetlands Ecosystems (RWE) on Peatland and Tidal Wetlands, and (4) Avoiding Planned Wetland Degradation (APWD) on Peatland and Tidal Wetlands. It is estimated that the project will generate a total net Verified Carbon Unit (VCU) volume of 69,999,940 tCO₂e over 60 years of crediting period or 1,166,666 tCO₂e per annum (Table 3-44).

The project's goal is to conserve the remaining forest, restore ecosystem function, protect biodiversity, and improve the well-being of communities to become resilient to climate change. In conclusion, it is expected that the PLUM Project will contribute to preserving the integrity of the Peat Swamp and Mangrove Forest landscape and biodiversity in the area while improving the well-being of local communities.

ha of rice areas that were excluded from the project area in consultation with local stakeholders²⁷. This project area (23,153.75 ha) is where the project activities aim to generate net climate benefits. The project zone is defined as the project area plus the surrounding area where the project activities directly affect land and associated resources, including activities related to the provision of livelihood improvement and community development. This project zone covers 75,280.54 ha of land (Map 2-7). The project zone boundary was determined based on village administrative boundaries (13 villages) and exclusion of conservation/protection areas (e.g., national park, protection forest, other ecosystem restoration concession).



²⁷ The final license document refers to 23,665 ha. However, conservatively the area of 23,631.51 ha is used which refers to the approved coordinates for the principal license. The project area is 23,153.75 ha excluding 467.68 ha of SIMURP rice area and approximately 10 ha of rice area in the western part of the project.

Stakeholder	Description	Role in the Project
Katingan Hulu Unit XVII	forest areas or is dominated by production forest areas.	
Bogor Agricultural University (IPB)	State-owned agricultural university based in Bogor, Indonesia	Support – as academic institution
Palangkaraya University	State-owned university located in Palangkaraya, Central Kalimantan	Support – as academic institution
Borneo Nature Foundation	Not-for-profit wildlife and biodiversity conservation and research organization that protects and safeguards tropical rainforests and the environment in Borneo.	Support – as non-profit conservation organization
Dompot Dhuafa	Health focused NGO	Support – as non-profit organization
PT Best Agro	Oil palm company	Business
PT Katingan Mujur Sejahtera	Oil palm company	Business
PT Katingan Sawit Bersama	Oil palm company	Business
PT Persada Era Agro Kencana	Oil palm company	Business
PT Pulang Pisau Antang Perkasa	Oil palm company	Business
PT Trubus Bumi Sejahtera	Oil palm company	Business

2.1.10 Sectoral Scope and Project Type

This project belongs to Sectoral Scope 14 for 'Agriculture, Forestry, and Other Land-Use (AFOLU)'. The project comprises REDD (APD) and ARR activities on peatland and tidal wetland as defined in VM0007 (Version 1.6). PLUM Project activities covered by the VM0007 include (1) Avoiding Planned Deforestation (APD), (2) Afforestation, Reforestation and Revegetation (ARR), (3) Restoration of Wetlands Ecosystems (RWE) on Peatland and Tidal Wetlands, and (4) Avoiding Planned Wetland Degradation (APWD) on Peatland and Tidal Wetlands.

The project is not a grouped project.

2.1.11 Project Activities and Theory of Change (G1.8)

The project aims to reduce emissions from planned deforestation and forest degradation while providing tangible climate, community, and biodiversity benefits. The area is directly threatened by the potential conversion to commercial oil palm plantation, which was successfully halted by the project. A portion of the project area is in a degraded condition due to historical logging, peatland drainage, and forest fires. Low community welfare surrounding the project area due to various causes, such as limited livelihood opportunities and low land productivity, have also indirectly threatened the integrity of the area through community agriculture expansion and illegal logging.

In the baseline scenario, the existence of commercial oil palm plantation in the area may provide some livelihood opportunities for the communities. However, it will not address the other issues currently faced by the communities, such as the limited access to health care services and education, limited livelihood opportunities and marketing of agricultural products, climate change, land degradation, and low water

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology

The methodology title and reference are the approved VCS Methodology VM0007 REDD+ Methodology Framework (REDD+ MF), v.1.6, 8 September 2020.

The tools applied by the project are: 1) the VT0001 Tool for the Demonstration and Assessment of Additionality in AFOLU Project Activities, v.3.0, 1 February 2012 (T-ADD); 2) the Agriculture, Forestry, and Other Land Use (AFOLU) Non-Permanence Risk Tool (NPRT), v.4.2, 12 October 2023 (T-BAR); 3) Tool for testing significance of GHG emissions in A/R CDM project activities (T-SIG) v.0.1.

3.1.2 Applicability of Methodology

The project applies the latest version of VCS Methodology VM0007 (Version 1.6 dated 08 September 2020) also called 'REDD+ MF' together with applicable pre-framed modules and tools. The modules were selected based on the PLUM Project activities covered by the VM0007 including (1) Avoiding Planned Deforestation (APD), (2) Afforestation, Reforestation and Revegetation (ARR), (3) Restoration of Wetlands Ecosystems (RWE) on Peatland and Tidal Wetlands, and (4) Avoiding Planned Wetland Degradation (APWD) on Peatland and Tidal Wetlands. All applicability conditions of the VM0007 (REDD+ MF) and its associated modules/tools are met by the project.

Table 3-1. Analysis of applicability conditions for the VM0007 methodology.

No	Module	Applicability Condition	Comment
1	REDD+ MF Version 1.6, Section 4.2, All Project Activities	All land areas registered under the CDM or under any other GHG program (both voluntary and compliance oriented) must be transparently reported and excluded from the project area. The exclusion of land in the project area from any other GHG program must be monitored over time and reported in the monitoring reports.	Condition met, no part of the project area is under any CDM or any other GHG program (either voluntary or compliance oriented).
2	REDD+ MF Version 1.6, Section 4.3.1, All REDD Activity Types	Land in the project area has qualified as forest (following the definition used by VCS) for at least the 10 years prior to the project start date.	Condition met. Land use records indicate that all land subject to REDD project activities in the project area is covered by tropical forest on peatland and has qualified as such under the applicable definition for at least 10 years.
3	REDD+ MF Version 1.6, Section 4.3.1, All REDD Activity Types	If land within the project area is peatland or tidal wetlands and emissions from the SOC pool is deemed significant, the relevant WRC modules (see Table 3) must be applied alongside other relevant modules.	Condition met. All relevant WRC modules have been applied to estimate emissions from peat soils.

No	Module	Applicability Condition	Comment
4	REDD+ MF Version 1.6, Section 4.3.1, All REDD Activity Types	Baseline deforestation and forest degradation in the project area fall within one or more of the following categories: Unplanned deforestation (VCS category AUDD), Planned Deforestation/ degradation (VCS category APD), Degradation through extraction of wood for fuel (fuelwood and charcoal production) (VCS category AUDD)	Condition met. Baseline deforestation falls in the category of APD. The project area does not include any significant unplanned deforestation or forest degradation and is not modelled in either the baseline of the project scenario.
5	REDD+ MF Version 1.6, Section 4.3.1, All REDD Activity Types	Leakage avoidance activities must not include: Agricultural lands that are flooded to increase production (e.g., rice paddy) Intensifying livestock production through use of feed lots 10 and/or manure lagoons.	Condition met. The project does not promote either establishment of agriculture on flooded land or intensification of livestock production.
6	REDD+ MF Version 1.6, Section 4.3.3, Avoiding Planned Deforestation/ Degradation	Avoiding planned deforestation/degradation activities are applicable under the following condition: Where conversion of forest lands to a deforested condition must be legally permitted.	Condition met. See baseline scenario
7	REDD+ MF Version 1.6, Section 4.4, ARR	The project area is non-forest land or land with degraded forest.	Condition met. See baseline scenario
8	REDD+ MF Version, Section 4.4, ARR	In strata with drained organic soil, ARR activities must be combined with rewetting.	Condition met. The project scenario includes rewetting activities.
9	REDD+ MF Version 1.6, Section 4.4, ARR	ARR activities are not eligible if the project scenario involves the application of nitrogen fertilizers.	Condition met. The project does not involve the application of any fertilizers.
10	REDD+ MF Version 1.6, Section 4.4, ARR	Where project activities on wetlands are excluded by the applicability conditions of applied modules or tools, these can be disregarded for the purpose of their use within this methodology, as quantification procedures for the peat soil are provided in modules BL-PEAT and M-PEAT.	Condition met. The project applies modules BL-PEAT and M-PEAT alongside all modules related to ARR.
11	REDD+ MF Version 1.6, Section 4.5.1, All WRC Activity Types	WRC activities are not eligible under the following conditions: > Project activities lower the water table, unless the project converts open water to tidal wetlands, or improves the hydrological connection to impounded waters. > Changes in hydrology do not result in the accumulation or maintenance of	Condition met. The project activities do not lower the water table, hydrological connection of the peatland does not lead to increase of emissions outside of the project area and do not include burning of organic soil, and does not include the use of fertilizers.

No	Module	Applicability Condition	Comment
		SOC stock, noting that a) this pertains to projects that intend to sequester carbon through sedimentation and/or vegetation development and b) this does not pertain to projects that increase salinity to reduce CH₄ emissions. > Projects that aim to decrease CH₄ emissions through increased salinity must account for any changes in SOC stocks. > Hydrological connectivity of the project area with adjacent areas leads to a significant increase in GHG emissions outside the project area. > Project activities include the burning of organic soil. Nitrogen fertilizer(s), such as chemical fertilizer or manure, are applied in the project area during the project crediting period.	
12	REDD+ MF Version 1.6, Section 4.5.2, RWE Project Activities General	For RWE project activities, prior to the project start date, the project area must meet the following conditions: The area is free of any land use that could be displaced outside the project area, as demonstrated by at least one of the following, where relevant: • The project area has been abandoned for two or more years prior to the project start date; or • Use of the project area for commercial purposes (i.e., trade) is not profitable as a result of salinity intrusion, market forces, or other factors. In addition, timber harvesting in the baseline scenario within the project area does not occur; or • Degradation of additional wetlands for new agricultural/aquacultural sites within the country will not occur or is prohibited by enforced law. OR b) The area is under a land use that could be displaced outside the project area, although in such case, baseline emissions from this land use must not be accounted for, and where degradation of additional wetlands for new agricultural/aquacultural sites within the country will not occur or is prohibited by enforced law. OR	Condition met. The project will continue at a similar or greater level of service or production during the project crediting period as the concession is under a Forest Utilization Business License (PBPH). The license is valid until the end of the project crediting period. In line with the granted license, the project will develop a long-term and annual management plan (RKU/RKT) which will include restoration of wetland ecosystem in the project area.

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation

Data / Parameter	$\Delta C_{BSL,planned}$
Data unit	t CO ₂ e
Description	Net greenhouse gas emissions in the baseline from planned deforestation
Equation	3, 14
Source of data	Module BL-PL
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	See Module BL-PL, among others the module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	$\Delta C_{BSL-ARR}$
Data unit	t CO ₂ e
Description	Net GHG removals in the ARR baseline scenario up to year t*
Equation	5
Source of data	Module BL-ARR
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	See Module BL-ARR, among others the project scenarios involving the harvesting of trees are excluded from this module.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	GHG _{BSL-PEAT}
Data unit	t CO ₂ e
Description	Net GHG emissions in the WRC baseline scenario up to year t*
Equation	6, 7
Source of data	Module BL-PEAT
Value applied	N/A

Justification of choice of data or description of measurement methods and procedures applied	See Module BL-PEAT, among others this module applies to the baseline scenario of WRC project activities on peatlands that are expected to be or remain (partly) drained in the absence of the project activity.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	GHG _{BSL-TW}
Data unit	t CO ₂ e
Description	Net GHG emissions in the WRC baseline scenario up to year t*
Equation	6, 7
Source of data	Module BL-TW
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	See Module BL-TW, among others this module applies to the baseline scenario of WRC project activities on tidal wetlands that are expected to be or remain (partly) drained in the absence of the project activity.
Purpose of data	Calculation of baseline emissions
Comments	N/A

3.3.2 Data and Parameters Monitored

Data/Parameter	$\Delta C_{WPS-REDD}$
Data Unit	t CO ₂ e
Description	Net GHG emissions in the REDD project scenario up to year t*
Equation	2
Source of data	Module M-REDD
Description of measurement methods and procedures to be applied	Applicable to monitoring ex-post emissions and removals of GHGs due to avoiding planned deforestation and forest degradation, and carbon stock enhancement that has been induced as a result of REDD project implementation within the project area and leakage belt and as a result of natural disturbances
Frequency of monitoring/recording	See Module M-REDD (Every 10 years (when the project baseline must be revisited) or every five years where conditions trigger more frequent baseline renewal)
Value applied	N/A
Monitoring equipment	Tool T-SIG
QA/QC procedures to be applied	See Module M-REDD
Purpose of Data	Calculation of project emissions
Calculation method	See Module M-REDD

Comments	N/A
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Data/Parameter	$\Delta C_{WPS-ARR}$
Data Unit	t CO ₂ e
Description	Net GHG emissions in the ARR project scenario up to year t*
Equation	5
Source of data	Module M-ARR
Description of measurement methods and procedures to be applied	See Module M-ARR
Frequency of monitoring/recording	See Module M-ARR
Value applied	N/A
Monitoring equipment	See Module M-ARR
QA/QC procedures to be applied	See Module M-ARR
Purpose of Data	Calculation of project emissions
Calculation method	See Module M-ARR
Comments	N/A

Data/Parameter	GHG _{WPS-PEAT}
Data Unit	t CO ₂ e
Description	Net GHG emissions in the WRC project scenario up to year t*
Equation	6
Source of data	Module M-PEAT
Description of measurement methods and procedures to be applied	See Module M-PEAT
Frequency of monitoring/recording	See Module M-PEAT
Value applied	N/A
Monitoring equipment	See Module M-PEAT
QA/QC procedures to be applied	See Module M-PEAT
Purpose of Data	Calculation of project emissions
Calculation method	See Module M-PEAT
Comments	N/A

Data/Parameter	GHG _{WPS-TW}
Data Unit	t CO ₂ e
Description	Net GHG emissions in the WRC project scenario up to year t*
Equation	6
Source of data	Module M-TW

Description of measurement methods and procedures to be applied	See Module M-TW
Frequency of monitoring/recording	See Module M-TW
Value applied	N/A
Monitoring equipment	See Module M-TW
QA/QC procedures to be applied	See Module M-TW
Purpose of Data	Calculation of project emissions
Calculation method	See Module M-TW
Comments	N/A

Data/Parameter	Frequency of patrol activities
Data Unit	Number of activities per year
Description	Record of the number of patrol activities carried out in the project area and leakage belt during the monitoring period.
Equation	N/A
Source of data	Patrol reports
Description of measurement methods and procedures to be applied	To be established
Frequency of monitoring/recording	To be established
Value applied	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Calculation method	N/A
Comments	N/A

Data/Parameter	Monitoring of forest cover by high-resolution satellite imagery
Data Unit	Number of activities per year
Description	Presentation of monitoring reports on land cover and land cover changes through high resolution satellite images.
Equation	N/A
Source of data	Forest cover monitoring reports
Description of measurement methods and procedures to be applied	High-resolution satellite data obtained through reliable and valid data sources, will be analyzed in order to examine the forest coverage monitoring data in the project area and leakage belt. To identify changes in land use in the monitored area, the data from the examined periods will be automatically categorized and visually interpreted.

Frequency of monitoring/recording	To be established
Value applied	N/A
Monitoring equipment	Geographical Information System (GIS) and Remote Sensing (RS) including the variety of tools, will be selected to support monitoring processes
QA/QC procedures to be applied	Precision target of >90% confidence interval at +/-10% of mean. >90% [Overall Classification Accuracy = 'Number of Pixels Classified Correctly' divided by 'Total Number of Classified Pixels']. Resolution (satellite imagery): <= 30 meter (m). Selection of Remote sensing instruments are, amongst others: Landsat, Resourcesat-1, Sentinel-1, Sentinel-2, JERS-1, ALOS/PALSAR, SPOT, Rapid Eye, PlanetLab. Cloud cover <10% during 12-month period.
Calculation method	N/A
Comments	N/A

3.3.3 Monitoring Plan

Climate impacts will be monitored in accordance with Appendix 13: Monitoring Plan, which includes land cover change monitoring, fire monitoring (e.g., hotspots, burnt areas), peat water table and subsidence monitoring following the requirements under VCS VM0007 methodology. The project will use both remote sensing approach and field data measurement for monitoring climate impacts. The monitoring plan outlines the indicators to be monitored, monitoring frequency, and data collection method.

3.3.3.1 Remote Sensing

Remote sensing approach will be used to monitor land cover change and burnt areas (if any fire incidences occur within the project area) and will be conducted annually. Satellite imageries with resolution of ≤ 30 meters (e.g., PlanetLab, Sentinel-1, Sentinel-2, ALOS/PALSAR, etc.) with cloud cover <10% will be used. Land cover change analysis will follow the procedures set out in Appendix 1: Remote Sensing Approach and Land Cover Classification Reports. Any deforested areas within the project area and project zone will be groundtruthed by the field team by directly visiting the location and/or using drones if the location is inaccessible. Overall classification accuracy is targeted at >90%.

Leakage monitoring in this project is undertaken to address the risk that deforestation activities are displaced elsewhere. By focusing on a specific identified actor, namely PT AIS, the project applies a targeted approach that draws on historical information and present circumstances and practices to demonstrate low potential for project-induced leakage.

Additionally, a robust and diversified monitoring strategy is implemented by the project as a commitment to transparency and environmental integrity, as follows:

- Remote Sensing: Satellite imagery tracking land cover changes across the project area and project zone to provide a general view of possible changes in land use patterns indicative of leakage.

- **GLAD Alerts:** The incorporation of the Global Land Analysis & Discovery system to provide near-real-time deforestation alert data, a tool that helps in rapid identification of unexpected changes to forest cover and enables prompt investigation and response.
- **Ground Truthing:** Regular field visits shall be undertaken to verify remote sensing and alert data to ensure accuracy for detection and attribution of leakage to actual activities or developments.

These sources of information are gathered and analyzed each time a new verification event occurs, with a special focus on the production, changes in the water table, and new oil palm developments. This integrated approach not only meets the specifications of the VMD0009 LK-ASP module but also serves to further build the project's capacity in monitoring any potential leakage and responding appropriately, hence delivering on its commitment to reducing deforestation and the related climate impact.

The project's approach to monitoring tidal wetlands adopts a comprehensive strategy to effectively curb GHG emissions. By combining direct measurements of GHG flux through both chamber methods with water quality evaluations for nutrients and dissolved carbon, this method offers a robust analysis of environmental health. The use of remote sensing technology, such as satellite imagery and drone surveys, provides detailed insights into the dynamics of the wetland, including changes in vegetation and shoreline changes. Ground evaluations further contribute by assessing the vegetation and soil's carbon storage potential. Monitoring the hydrological aspects, such as water table levels and the impact of tides, is fundamental for understanding how water movements influence emission trends. This integrated system ensures that the project can enhance the wetland's capacity for carbon capture while keeping GHG releases to a minimum.

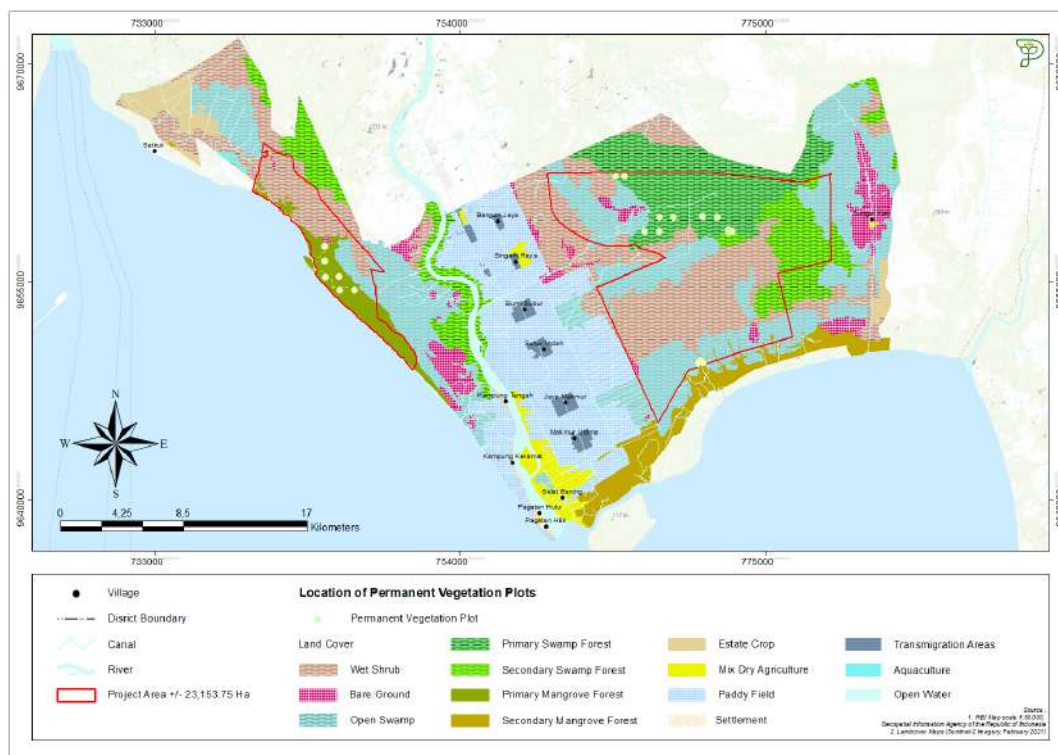
3.3.3.2 *Field Data Measurement*

3.3.3.2.1 *AGB C Stock*

To monitor the C stock within the project area, the project will establish 20 permanent vegetation/biomass plots in the same locations that were used for the baseline assessment. The plots will be monitored annually for tree growth (DBH). To estimate the C stock from DBH, we will use the same allometric equations that were used when determining the baseline AGB C stock⁷⁴.

For ARR activities, the project will establish new plots that will represent the areas of ARR (including ANR) activities for both mangrove and peat ARR. DBH will be the main parameter measured in the field and species specific allometric equations will be used to estimate the C stock (in accordance with the species that will be planted). Field measurement will be conducted annually. In addition, the project will create a logbook that records the number of seedlings planted by species, survival rate, DBH, and height for active planting activities.

⁷⁴ PT PUM's Biomass Carbon Stock Assessment Report.



Map 3-10. Planned location of permanent vegetation plots

3.3.3.2.2 Fire Hotspots and Burnt Areas

The project will develop a Fire Danger Rating System (FDRS) to serve as an early warning system of fire risk in the project area. Automatic Weather Stations (AWS) will be installed on the ground to collect climate data that correlate with fire risk (e.g., temperature, precipitation, soil moisture, and peat water level) and these data will be supplied into the FDRS to calculate the Peat Fire Vulnerability Index (PFVI). The FDRS will then provide a 7-day forecast of fire risk in the project area. The FDRS will categorize the fire risk based on PFVI's value as Safe / Alert / Dangerous / Extreme.

In the field, the project will monitor fire hotspots through daily monitoring from fire watch tower and fire prevention patrols during the dry season when the FDRS rates the fire risk as Alert / Dangerous / Extreme. In addition, the project also monitors fire hotspots from NASA's Fire Information for Resource Management System (FIRMS). FIRMS provides near real-time active fire data within three hours of satellite observation from the MODIS and VIIRS instruments. The fire patrol team will verify fire hotspots on the ground by directly visiting the location and/or using drone if the location is inaccessible.

Burnt areas within the project area will be mapped and calculated using remote sensing approach with high-resolution satellite imagery within one month of fire incidences with detailed procedure listed in the Standard Operational Procedure document for Prevention and Management of Forest and Land Fire (please refer to Appendix 12: List of SOPs). Areas with repeated burning will be analyzed using historical burning records. GHG emissions from biomass and peat burning will be calculated annually following the E-BPB module.

3.3.3.2.3 Peat Water Table and Subsidence

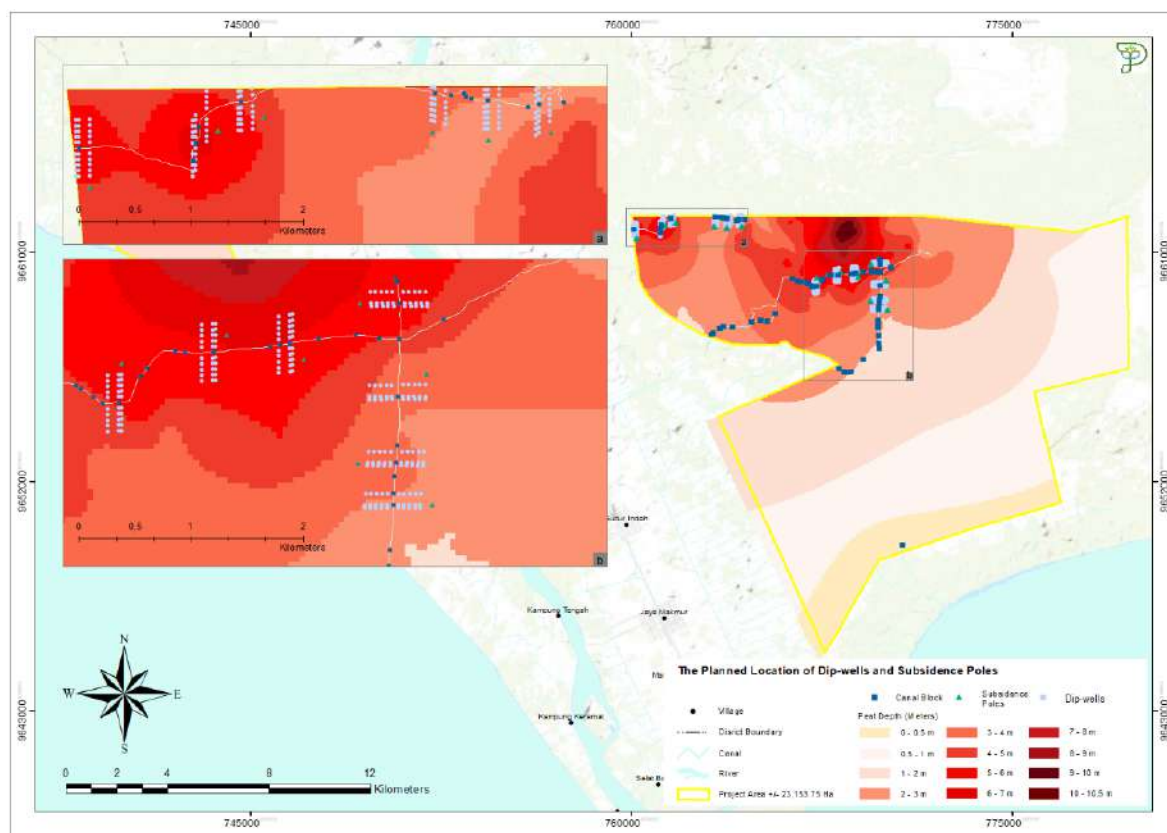
The project plans to install 352 dipwells and 13 subsidence poles along the planned canal blocks in the project area to monitor peat water level and peat subsidence. However, the numbers of monitoring stations may be adjusted depending on field conditions. Peat water level will be measured automatically (daily) or manually (weekly), while peat subsidence will be monitored manually every three months.

The project implements a detailed approach to monitor peat water levels and subsidence, crucial for evaluating its impact on peatland conservation and carbon sequestration. Through both automatic (daily) and manual (weekly) measurements of water levels, as well as quarterly manual checks of peat subsidence, it captures essential data on hydrology and land stability. This is followed by data analysis on peat water levels and subsidence to provide further understanding of this influential factor on greenhouse gas (GHG) emissions. Water-level fluctuations in peat, in fact, determine the rate of decomposition. Generally, low water levels imply more decomposition, hence higher emissions of GHGs. Equally, peat subsidence data provide information on the long-term stability in the peat and also the storage of carbon in peatland. Summed with other data points, the project can then very accurately infer GHG emissions from peat decomposition in tropical peatlands.

3.3.3.2.4 GHG Emissions

The project is exploring future installation of greenhouse gas (GHG) monitoring equipment aimed at providing long-term data on atmospheric GHG flux in peatlands, including carbon dioxide (CO₂), methane (CH₄), among others. Atmospheric monitoring of GHG flux enhances the understanding on the sequestration of carbon from the atmosphere by vegetation and its subsequent integration into the soils. The project may leverage atmospheric-based monitoring, potentially employing GHG flux towers, in collaboration with Hyphen Global AG to further these objectives.⁷⁵

⁷⁵ Hyphen Global AG is a Switzerland-based climate tech company pioneering science driven digital MRV by way of precision emissions validation through real-time atmospheric greenhouse gas (GHG) monitoring. Atmospheric-based methodologies for monitoring these processes receive international endorsement from authoritative bodies such as the Intergovernmental Panel on Climate Change (IPCC), the United Nations Framework Convention on Climate Change (UNFCCC), and the World Meteorological Organization (WMO). These endorsements underscore the peer-reviewed accuracy of measured sequestration data obtained through atmospheric monitoring.



Map 3-11. Planned location of canal blocks, dip-wells and subsidence poles

Table 3-45 Monitoring Indicators

No	Monitoring Indicator	Remote Sensing	Field Data	Monitoring Frequency
1	Land Cover Change	●	●	Annually
2	Shoreline Change	●	●	Annually
3	AGB C Stock		●	Annually
4	Fire Hotspots and Burnt Areas	●	●	Daily during the dry season, burnt areas analysis within one month of incidence
5	Peat Water Table		●	Daily (automatic)/ weekly (manual)
6	Peat Subsidence		●	Every three months

3.3.3.3 Data Management

The procedures for data management, internal auditing and QA/QC are described in Figure 3-7 below. Please also refer to Section 2.3.8 and Figure 2-4 on the project's adaptive management.

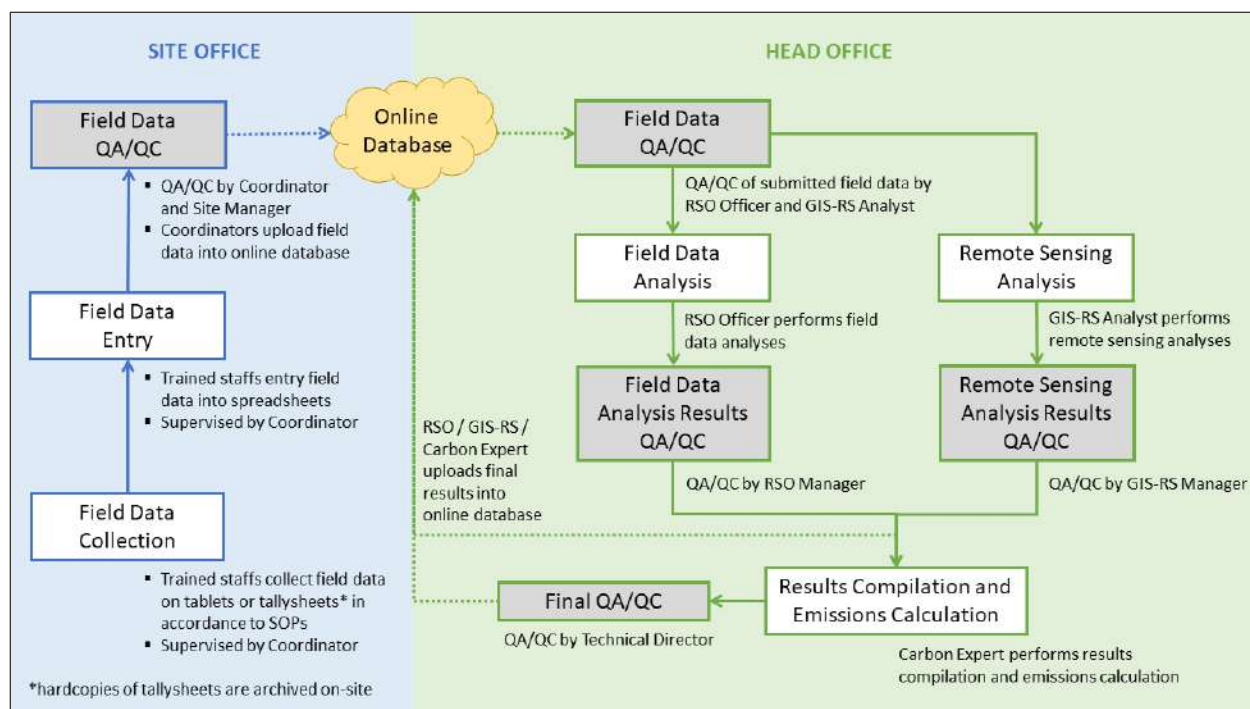


Figure 3-7. Data management process

The project will also adopt the Spatial Monitoring and Reporting Tool (SMART) Patrol System⁷⁶ to digitize the process of field data collection (including data for Community and Biodiversity) and therefore minimizing the risk of human error in manual data entry. The SMART Patrol Mobile application will be installed on tablets or smartphones that will be used by the field staff for data collection. The SMART Patrol Desktop application will serve as the main database of all field data.

To avoid non-conformances with the monitoring plan, the project has developed and will develop SOPs at each level (e.g., field data collection, data management, data analysis, etc.). A list of available SOPs and planned SOPs that will be developed is available in Appendix 12: List of SOPs. A copy of SOPs can be provided to the validator upon request. All project staff will be trained to follow procedures in accordance with the SOPs and will continuously be supervised. The project will also implement multiple QA/QC at various levels. All of the data collected, entered, uploaded, and analyzed are subject to review and approval by the Coordinator or Manager or Expert or Director (depending on the level) to ensure that the SOPs are correctly followed (Figure 3-7). In addition, the project also adopts adaptive management principle (please refer to Section 2.3.8) and will regularly evaluate the implementation of monitoring procedures and make improvements when required.

The following table provides the list of monitoring team members, responsibilities, and competencies.

⁷⁶ <https://smartconservationtools.org/>

Table 3-46. Monitoring team members

No	Name / Position	Responsibilities	Competencies
1	TBD / Technical Director	Oversee and provide technical advice on all processes, final QA/QC of technical analysis and emissions calculation	Please refer to Section 2.1.4
2	TBD / GIS-Remote Sensing Manager	Supervise SOPs development, remote sensing and GIS analyses QA/QC	Please refer to Section 2.1.4
3	Imanuddin Utoro / RSO Manager	Supervise SOPs development, field data analysis QA/QC	Please refer to Section 2.4.3
4	TBD / Carbon Expert	Develop carbon models, calculate emissions from monitoring data analyses results	Please refer to Section 2.1.4
5	Vanny Milenia / GIS-Remote Sensing Analyst	Development of SOPs, spatial field data QA/QC, perform remote sensing and GIS analyses	Please refer to Section 2.4.3
6	Larissa D Salaki / Senior RSO Officer	Development of SOPs, field data QA/QC, field data analysis, progress tracking	Please refer to Section 2.4.3
7	Andreina Triyantari / RSO Officer	Development of SOPs, field data QA/QC, field data analysis, progress tracking	Please refer to Section 2.4.3
8	Siti Nadia / RSO Officer	Development of SOPs, field data QA/QC, field data analysis, progress tracking	Please refer to Section 2.4.3
9	Soni Budiawan / Site Manager	Field data QA/QC	Please refer to Section 2.4.3
10	Pandji A. Fauzan / Biodiversity & Forest Protection Coordinator	Assist and supervise field staff on field data collection and entry, field data QA/QC, field data upload	Please refer to Section 2.4.3
11	Field Staff	Field data collection, field data entry	Various, please refer to Section 2.4.1

RSO = Restoration and Sustainability

3.3.4 Dissemination of Monitoring Plan and Results (CL4.2)

There are three types of stakeholders involved in the project, namely government, local communities, and business entities. For all stakeholders, the summary of project documents, including monitoring plan and results, will be shared and given access to through regular coordination, printed documents at the project offices and village information boards, and online resources through Verra and the project website (www.plumproject.id).

Specifically for communities, the project documents will be shared through village meetings and village information boards. Regular communication will occur during the project lifetime, as the project will closely involve the community in the project activities and monitoring. Formally, at least two village meetings will be conducted annually to disseminate project activities updates and reports.